

RootWork Learning Hub — Production Readiness Assessment

Date: 2026-02-07 **Version:** 0.1.0 **Assessed by:** Automated analysis of repository SAHearn1/learning-hub

Executive Summary

The RootWork Learning Hub has a **solid architectural foundation** but is currently in **early Phase 1** — the scaffolding, data modeling, and configuration layers are in place, while the core functional features (AI tutoring sessions, assessments, progress tracking, payment flows) remain unimplemented. The project is **not production-ready** and requires significant implementation work across 7 critical areas before students and teachers can use it.

Overall readiness: ~20% toward MVP launch.

1. Build & Compilation Status

Check	Result
next build	PASS — 21 routes compiled, zero errors
next lint	PASS — no warnings or errors
TypeScript strict mode	Enabled, compiles cleanly
Prisma schema validation	PASS (requires DATABASE_URL at runtime)
npm audit	5 moderate vulnerabilities (all in dev-only vitest / vite / esbuild chain)

Verdict: The codebase compiles and lints cleanly. No blockers here.

2. What Is Built (Phase 1 Complete)

These foundational layers are in place and well-structured:

- **Authentication & middleware** — Clerk integration with route protection, role-based auth helpers (`getCurrentUser` , `requireUser` , `requireRole`)
- **Database schema** — 30+ Prisma models covering multi-tenancy, users/roles, sessions, assessments, curriculum standards, IEP accommodations, AI usage tracking, audit logging
- **State management** — 5 Zustand stores (session, progress, regulation, learner, UI) with clear interfaces
- **AI configuration** — Anthropic client initialized, dual-model setup (Sonnet + Haiku), 100+ line master system prompt for the "RootGuide" trauma-informed tutor persona
- **Type system** — Comprehensive TypeScript interfaces for sessions, learners, assessments, curriculum, API responses
- **Configuration layer** — Feature flags, pricing tiers, navigation by role, subject-grade mappings, 5Rs phase transitions, permission matrix
- **Page scaffolding** — 19 routes with stub content across Student, Educator, Parent, and Admin portals
- **Security headers** — X-Frame-Options, X-Content-Type-Options, Referrer-Policy, Permissions-Policy
- **Stripe initialization** — Client configured (no checkout or webhook handling yet)

3. Critical Gaps — What Is NOT Built

3.1 API Routes (ZERO exist)

No `src/app/api/` routes are implemented. This means:

- No AI tutoring conversation endpoint
- No session create/read/update/delete
- No assessment generation or submission
- No progress retrieval
- No Stripe checkout session creation
- No Stripe webhook handler
- No user onboarding/profile sync with Clerk
- No educator data retrieval endpoints

- No parent data retrieval endpoints

Priority: CRITICAL — nothing works without API routes.

3.2 Core Tutoring Flow (Learn Page)

The `/learn` page is a placeholder with text: *"Guided lesson flow will live here in Phase 2.1."*

Required implementation: - Chat interface with message input/display - Real-time streaming from Anthropic API - 5Rs phase transitions during sessions - Engagement mode switching (Forward, Reverse, Error Analysis, Multiple Pathways, Problem Posing) - Dysregulation signal detection and Calm Corner trigger - Session persistence to database - TRACE protocol integration

3.3 Assessment System

No assessment UI or logic exists: - Diagnostic placement assessments - Formative in-session checks - Summative mastery evaluations - Thinking quality and creativity scoring - Reasoning move tracking (21 move types)

3.4 Progress Dashboard

The `/progress` page is a stub. Needs: - Mastery visualization by standard - Session history - Reasoning move growth charts - Bloom's taxonomy progression - Export/print for parents and educators

3.5 Educator Portal Functionality

All educator pages (`/educator/students` , `/educator/classes` , `/educator/reports` , `/educator/compliance`) are stubs. Needs: - Student roster management - Class creation and enrollment - Progress reports with filtering - IEP accommodation management - Compliance reporting

3.6 Payment Processing

Stripe client is initialized but nothing else: - No checkout session creation - No subscription management - No webhook handler for payment events - No usage limit enforcement based on tier - No billing portal link

3.7 Database Not Provisioned

- No Prisma migrations have been created (`prisma/migrations/` does not exist)
- No seed script implementation (`prisma/seed.ts` referenced but not verified)
- No database provisioned for any environment

4. Testing — Current State: NONE

Test Type	Framework Installed	Tests Written	Coverage
Unit tests	Vitest 2.0.0	0	0%
E2E tests	Playwright 1.45.0	0	0%
Integration tests	—	0	0%

Required before production: - Unit tests for all API routes, auth helpers, state stores - Integration tests for tutoring conversation flow - E2E tests for student sign-up → session → progress flow - E2E tests for educator class management flow - Accessibility testing (critical for IEP-accommodated students) - Load testing for concurrent AI sessions

5. DevOps & Deployment — Current State: NONE

Requirement	Status
Dockerfile	Missing
docker-compose	Missing
CI/CD pipeline (.github/workflows)	Missing
Deployment config (Vercel/Railway/etc.)	Missing
Environment variable management	<code>.env.example</code> only
Database hosting	Not configured
Error monitoring (Sentry, etc.)	Not integrated
Logging infrastructure	Not configured
CDN/asset optimization	Not configured
SSL/TLS	Depends on deployment platform
Health check endpoint	Missing

6. Compliance & Legal Readiness

Given this is a K-12 education platform serving minors:

Requirement	Status	Notes
COPPA compliance	Partial	Consent model exists in schema, no enforcement flow
FERPA compliance	Partial	Audit logging in schema, no implementation
Privacy policy	Draft	<code>/privacy</code> page exists as stub
Terms of service	Draft	<code>/terms</code> page exists as stub
Data retention policy	Missing	No data lifecycle management
Parental consent flow	Missing	Schema has <code>ConsentStatus</code> , no UI/API
Data export/deletion (GDPR)	Missing	No user data export capability
SOC 2 readiness	Missing	No evidence of compliance controls

Priority: HIGH — serving minors requires robust consent and data protection.

7. Security Audit

Area	Status	Detail
Auth middleware	Good	Clerk middleware protects non-public routes
Security headers	Good	X-Frame-Options, CSP basics, etc.
API input validation	N/A	No API routes exist yet
Rate limiting	Missing	No rate limiting on any endpoint
CSRF protection	Partial	Server Actions enabled; needs review
Secrets management	Basic	<code>.env.example</code> documented; <code>.env</code> in <code>.gitignore</code>
Dependency vulnerabilities	Low risk	5 moderate (dev-only, vitest chain)
SQL injection	Low risk	Prisma ORM parameterizes queries

XSS protection

Area	Status	Detail
	Needs review	React escapes by default; need review of <code>dangerouslySetInnerHTML</code> usage

8. Production Readiness Scorecard

Category	Weight	Score (0-10)	Weighted
Build & compilation	5%	9	0.45
Core feature completeness	25%	1	0.25
API layer	20%	0	0.00
Testing	15%	0	0.00
DevOps & deployment	10%	0	0.00
Security	10%	5	0.50
Compliance (COPPA/FERPA)	10%	2	0.20
Documentation	5%	3	0.15
Total	100%		1.55 / 10

9. Recommended Path to Production — Phased Roadmap

Phase 2A: Core API & Tutoring MVP

1. **Database provisioning** — Create Prisma migrations, provision PostgreSQL (e.g., Supabase, Neon, Railway)
2. **Clerk webhook** — Sync user creation/updates to the `User` table
3. **API route: `/api/sessions`** — Create, retrieve, end tutoring sessions
4. **API route: `/api/chat`** — Streaming AI conversation endpoint using Anthropic SDK
5. **Learn page implementation** — Chat UI, message streaming, phase display
6. **Regulation detection** — Dysregulation signal matching in conversation
7. **Session persistence** — Save messages and session state to database

Phase 2B: Assessment & Progress

1. **API route:** `/api/assessments` — Generate and grade assessments
2. **Progress tracking API** — Mastery level computation and storage
3. **Progress dashboard UI** — Visualize mastery by standard, session history
4. **Thinking assessment integration** — Reasoning move tracking per session

Phase 2C: Educator & Parent Portals

1. **API routes:** `/api/educator/*` — Student roster, class management, reports
2. **Educator UI completion** — Functional student list, class management, compliance views
3. **Parent portal API & UI** — Child progress visibility, notification preferences
4. **IEP accommodation management** — CRUD for accommodations tied to student profiles

Phase 2D: Payments & Limits

1. **Stripe checkout** — Plan selection → Stripe Checkout → subscription creation
2. **Stripe webhooks** — Handle `checkout.session.completed`, `invoice.paid`, `subscription.updated`, `subscription.deleted`
3. **Usage enforcement** — Token/session limits per subscription tier
4. **Billing portal** — Link to Stripe Customer Portal for self-service management

Phase 3: Testing & Hardening

1. **Unit tests** — API routes, auth helpers, store logic (target: 80%+ coverage)
2. **E2E tests** — Full student and educator flows with Playwright
3. **Accessibility audit** — WCAG 2.1 AA compliance, screen reader testing
4. **Load testing** — Concurrent session capacity planning
5. **Rate limiting** — Implement per-user/per-IP limits on AI and API endpoints
6. **Error monitoring** — Integrate Sentry or equivalent

Phase 4: Deployment & Compliance

1. **CI/CD pipeline** — GitHub Actions for lint, test, build, deploy
2. **Deployment configuration** — Vercel (recommended for Next.js) or containerized deployment
3. **Database migrations workflow** — Automated migration in CI/CD
4. **COPPA/FERPA compliance implementation** — Parental consent flow, data retention policies, export/deletion

5. **Privacy policy & ToS finalization** — Legal review of content
 6. **Monitoring & alerting** — Uptime, error rate, AI latency dashboards
 7. **Staging environment** — Separate environment for QA before production deploys
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10. Immediate Next Steps (Top 5 Priorities)

1. **Provision PostgreSQL and run initial migration** — Everything depends on a working database
 2. **Implement Clerk webhook to sync users** — Auth is in place but users never persist to the DB
 3. **Build the `/api/chat` streaming endpoint** — The core AI tutoring experience
 4. **Implement the Learn page chat UI** — The primary student-facing feature
 5. **Set up CI/CD with GitHub Actions** — Prevent regressions from the start
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This assessment reflects the state of the codebase as of commit `7f597f7` on 2026-02-07.